

Curriculum Vitae : Sungjun Hong

Work Authorization

J-2 Visa with EAD (valid until March 2027)

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PROFESSIONAL SUMMARY

Biomedical statistician and data scientist with 7+ years of experience developing and applying statistical and machine learning methods for complex, high-dimensional healthcare data. Led development of a deep learning-based pipeline integrating U-Net segmentation with predictive modeling for histologic disease activity in ulcerative colitis, and developed a radiomics-based feature selection framework applied to prediction models across multiple gastrointestinal malignancies. Doctoral research focused on machine learning-based propensity score methodology for causal inference in observational studies, with extensive simulation-based methodological evaluation. Extensive applied experience analyzing electronic health records, national administrative claims databases, and clinical trial data across collaborative research projects at Samsung Medical Center, spanning study design, survival analysis, longitudinal modeling, and reproducible analytical workflows. Proficient in R, SAS, and Python.

TECHNICAL SKILLS

Statistical Methodology & Predictive Modeling

- Development, validation and application of predictive models for clinical outcomes using clinical imaging, and administrative data
- Development and validation of causal inference methods, including machine learning-based propensity score frameworks for multiple treatment groups
- High-dimensional feature selection methodology (filter, wrapper, embedded methods), evaluated for robustness, redundancy, and clinical relevance
- Simulation study design for methodological evaluation and comparison
- Bootstrap-based validation and performance assessment of statistical and predictive models

Applied Biostatistical Analysis

- Design and analysis of observational studies using clinical and real-world healthcare data
- Regression modeling for clinical outcomes (linear, logistic, survival, GEE models)
- Risk factor analysis and outcome evaluation using large administrative and clinical datasets
- Model performance assessment, and calibration evaluation

TECHNICAL SKILLS (CONT.)**Programming, Reproducible Research & Data Management**

- R: statistical modeling, data wrangling, reproducible workflows, and data visualization
- SAS: data management, statistical procedures, and macro programming
- Python: data preprocessing, statistical analysis, machine learning, and deep learning model development
- Integration, cleaning, and management of multi-source datasets (EHR, claims, survey data)

Scientific Reporting & Research Documentation

- Preparation of statistical analysis plans, technical reports, and scientific manuscripts
- Development of tables, figures, and visualizations for publications and presentations
- Clear documentation and communication of analytical methods and findings

EMPLOYMENT**Samsung Medical Center**

Research Institute for Future Medicine

Team Leader, Medical AI Research Center (September 2023 – February 2025)

Concurrent Academic Appointments at Sungkyunkwan University School of Medicine:

- **Research Professor** (September 2024 – February 2025)
- **Postdoctoral Researcher** (September 2023 – August 2024)

Researcher, Research Institute for Future Medicine (November 2017 – August 2023)

EDUCATION

Ph.D. in Digital Health (08.2023) Sungkyunkwan University, South Korea

M.S. in Statistics (02.2018) University of Seoul, South Korea

B.S. in Statistics (02.2015) University of Suwon, South Korea

RESEARCH INTERESTS

- Predictive Modeling; Machine Learning; Healthcare AI
- Clinical Data Science; Biostatistics; Causal Inference; Real-World Evidence

SELECTIVE PUBLICATIONS

Main Author (*Co-first author, **Co-corresponding author)

- **Hong S.***, K. Kim*, J. Hwang, J. Hur, Y. Chang, H. Cha, S. J. Cho, J. Kim, H. Son, H. Lim, J.K. Ahn**, M. Kang** (2026) Sex-Specific Association of Drinking Habit and Alcohol Beverage Type with Serum Uric Acid in Health Population. *Journal of Korean Medical Science*, 41(2), e35.
- Hong S*, **S. Hong***, E. Oh, W.J. Lee, W.K. Jeong**, K. Kim** (2024) Development of A Flexible Feature Selection Framework in Radiomics-based Prediction Modeling: Assessment with Four Real-World Datasets. *Scientific Reports*, 14(1), 29297.
- Yu J.*, **S. Hong***, Y.C. Lee, K.H. Lee, I. Lee, Y. Seo, M. Kang, K. Kim, W.C. Cha, S.Y. Shin (2022) Stakeholders' Requirements of Artificial Intelligence for Healthcare in Korea. *Healthcare Informatics Research*, 28(2), 143-151
- **Hong S.**, S. Lee, J. Lee, W.C. Cha**, K. Kim** (2020) Prediction of Cardiac Arrest in the Emergency Department Based on Machine Learning and Sequential Characteristics: Model Development and Retrospective Clinical Validation Study. *JMIR Medical Informatics*, 8(8), e15932.

Co-Author in Collaborative Research on Big Data (Health-checkup, Health insurance claim, etc.)

- Park Y. J.*, S. Lee*, **S. Hong**, K. Kim**, J. Kim, J. Y. Kim, K.-M. Park, Y. K. On, S.-J. Park** (2025) Temporal Trends and Prognostic Impact of Pacemaker-Associated Heart Failure: Insights from a Nationwide Cohort Study. *Journal of Clinical Medicine*, 14(21), 7744.

Co-Author in Collaborative Research on AI, Machine Learning, Digital Healthcare

- Hwang S.*, Heo S.*, **S. Hong**, K.-H. Jung, W.C. Cha, J. Yoo (2026) Multimodal AI-Based 28-Day Mortality Prediction of Pneumonia Patients at ED Discharge: A Multicenter Study. *Scientific Reports*, 16, 11761.
- Lim CY., D.I. Choi, W.K. Jeong, Y.Y. Cho, **S. Hong**, S. Hong, K. Kim, J.-H. Kim (2025) Prediction of microsatellite-stable/epithelial-to-mesenchymal transition molecular subtype gastric cancer using CT radiomics and clinicopathologic factors. *European Journal of Radiology*, 185, 111990.
- Lee S., **S. Hong**, W.C. Cha**, K. Kim** (2020) Predicting Adverse Outcomes for Febrile Patients in the Emergency Department Using Sparse Laboratory Data: Development of a Time Adaptive Model. *JMIR Medical Informatics*, 8(3), e16117.

Co-Author in Other Collaborative Clinical Research

- Kim M.*, M. Cho*, **S. Hong**, J.H. Song, E.R. Kim, S.N. Hong, D.K. Chang, Y. Kim, J.E. Kim (2023) Weight Loss from Diagnosis of Crohn's Disease to One Year Post-Diagnosis Results in Earlier Surgery. *Scientific Reports*, 13(1), 21101.
- Kim Y.*, C.H. Kim*, H.Y. Lee*, S.H. Lee*, H.S. Kim, S. Lee, H. Cha, **S. Hong**, K. Kim, S.W. Seo, J.M. Sun, M.J. Han, J.S. Ahn, K. Park (2019) Comprehensive Clinical and Genetic Characterization of Hyperprogression Based on Volumetry in Advanced Non-Small Cell Lung Cancer Treated With Immune Checkpoint Inhibitor. *Journal of Thoracic Oncology*, 14(9), 1608-1618.

SELECTIVE RESEARCH PROJECTS**Collaborating Researcher**

- Investigating the Usefulness of Histological Activity Evaluation using Deep Learning in Patients with Ulcerative Colitis (Sep. 2023 – Dec. 2024, Jun. 2025 – Mar. 2026)
 - Samsung Medical Center (PI: Young-Ho Kim; Gastroenterology @ SMC)
 - Developed a segmentation model for multiple cell types and a prediction model for Nancy-grade activity in patients with UC (Python)
 - Conducted external validation using a multi-center dataset
- Converged Image-based Precision Oncology AI Solution for Gastrointestinal Malignancies (May 2021 – Aug. 2023)
 - Samsung Medical Center, Future Medicine 20*30 Project (PI: Woo Kyoung Jeong; Radiology (Gastric Cancer Center) @ SMC)
 - Designed and validated a clinical prediction model for radio-genomics
 - Developed a radiomic-based feature selection framework (R)
- A Study on Non-parametric Propensity Score Method in Unbalance Multiple Treatment Groups (Jun. 2021 – Feb. 2023)
 - National Research Foundation of Korea (No. 2021R1F1A1054806, PI: Kyunga Kim; Biomedical Statistics Center @ SMC)
 - Developed statistical methodologies for PS matching in multiple-treatment settings, and evaluated performance using both simulation studies and real-world datasets (R)
- A Comparative Study of the Correlation between Insulin Resistance or Diabetes and Dietary Patterns applying Machine Learning and Statistical Techniques (Sep. 2022 – Feb. 2023)
 - PI: Yoong Jung Yang; Department of Food and Nutrition @ DWU, Kyunga Kim; Biomedical Statistics Center @ SMC
 - The Korea national health and nutrition examination survey
 - Applied clustering methods to identify dietary intake patterns (R, Python)
- The Effect of CRT Upgrade to Overall Survival in Pacing-Associated Heart Failure Patients (Aug. 2020 – Dec. 2023)
 - Samsung Medical Center (PI: Seung Jung Park; Heart Vascular Stroke Institute @ SMC)
 - Designed and analyzed Korea National Health Insurance claims data
 - Conducted survival analysis to evaluate the incidence of heart failure following pacemaker implantation (R, SAS)
 - Performed subgroup analysis using propensity score matching and time-dependent modeling (time-varying covariates and coefficients)

STATISTICAL CONSULTING EXPERIENCE

- Department of Radiology and Center for Imaging Science, Samsung Medical Center (Oct. 2021 – Aug. 2023)
- Provided statistical consulting for 100+ clinical research projects, including:
 - **Model Design** (Sample Size, Power Analysis, Study Design, Data Collection, Missing Imputation, Imbalance Problem, Etc.)
 - **Statistical Analysis** (Bootstrap, Linear/Logistic Regression, Longitudinal Analysis, GEE, Meta Analysis, Imbalance Problem, Non-Inferiority test, Data Visualization, Variable selection, Etc.)
 - **Survival Analysis** (Cox Regression, PH Assumption, Time-dependent Analysis, Competing Risk, Adjusted Survival Curve, Hierarchical Modeling, Etc.)
 - **Predictive Modeling** (Model Development, Validation, Evaluation, Application, Nomogram, Scoring system, Cutoff Selection, Bootstrap, Etc.)
 - **Propensity Score Method** (Matching, Weighting, Sensitivity analysis)
 - **Radiomics Research** (Feature Redundancy, Relevance, Selection, Radiomics Scoring, Variable Definition, Agreement test, Subgroup Analysis)

PROFESSIONAL SKILLS

- Statistical Packages
 - Proficient in R, SAS, SPSS
- Programming Languages
 - Proficient in R, SAS, Python
- Dashboard Visualization
 - Shiny in R
 - Plotly & Dash in Python

CERTIFICATIONS

- Advanced Data Analytics Semi-Professional (2020)
- SAS Analytic
 - SAS Certified Statistical Business Analyst Using SAS 9: Regression and Modeling (2017)
- SAS Foundation
 - SAS Certified Advanced Programmer for SAS 9 (2017)
 - SAS Certified Base Programmer for SAS 9 (2016)
- MS Office
 - Computer Specialist in Spreadsheet & Database Level-I (2014)
- Survey Analyst, Junior (2013)

OTHER PROFESSIONAL EXPERIENCE**Guest Lecturer**

- Department of Digital Health, SAIHST, Sungkyunkwan University, South Korea (November 18, 2024; September 29, 2025)
 - AI introduction for Healthcare Services
 - ✓ Practice 1: Python programming (Basic)
- Department of Digital Health, SAIHST, Sungkyunkwan University, South Korea (November 6, December 4, 2024)
 - Methodology for Healthcare Predictive Modeling
 - ✓ Predictor (feature) selection (Nov. 6)
 - ✓ Internal validation (Dec. 4)
- Medical Informatic Center, Samsung Medical Center, South Korea (August 19, 20, 2024)
 - Python Programming
 - ✓ Data Preprocessing with numpy, pandas
 - ✓ Data Visualization with matplotlib, plotly
- School of Medicine, Sungkyunkwan University, South Korea (June 21, 28, 2024)
 - Computational Medicine
 - ✓ Machine Learning I (Regression, Classification, and Clustering Methods with Python)
 - ✓ Machine Learning II (Variational Inference with Python)
- School of Medicine, Sungkyunkwan University, South Korea (May 22, 2024)
 - Capstone of Medical Artificial Intelligence II
 - ✓ Prediction Modeling: Metrics for Multi-class Classification
- Department of Digital Health, SAIHST, Sungkyunkwan University, South Korea (November 7, 14, 2023)
 - Introduction to Medical Data Science
 - ✓ Unsupervised Learning – Clustering Methods
- College of Nursing, The Catholic University of Korea (March 23, 2022)
 - Nursing Research and Statistical Analysis I
- Seoul Metropolitan Fire & Disaster Headquarters, South Korea (January 25, 2022)
 - Application of Medical Big Data

Lecture Assistant

- Capstone of Medical Artificial Intelligence II, School of Medicine, Sungkyunkwan University, South Korea (1st semester, 2024)
 - Clinical Prediction Modeling
- Capstone of Medical Artificial Intelligence I, School of Medicine, Sungkyunkwan University, South Korea (2nd semester, 2023, 2024)
 - Deep Learning: Theory and Practice with Python (Pytorch)
- Undergraduate Teaching at Department of Statistics, University of Seoul, South Korea (2016 – 2017)
 - Applied Probability Theory
 - Matrix Algebra I, II

Full List of Publications

PUBLICATIONS

Papers at Refereed Journals in 2026 (*Co-first author, **Co-corresponding author)

- Hwang S.*, Heo S.*, S. Hong, K.-H. Jung, W.C. Cha, J. Yoo (2026) Multimodal AI-Based 28-Day Mortality Prediction of Pneumonia Patients at ED Discharge: A Multicenter Study. *Scientific Reports*, 16, 11761.
- Park Y.J.*, S. Lee*, S. Hong, K. Kim**, J. Kim, J.Y. Kim, K.-M. Park, Y.K. On, S.-J. Park** (2026) Long-Term Mortality and Treatment Outcomes in Pacemaker-Associated Heart Failure: Insights from a Nationwide Propensity-Matched Study. *European Heart Journal Open*, oeag027.
- Hong S.*, K. Kim*, J. Hwang, J. Hur, Y. Chang, H. Cha, S. J. Cho, J. Kim, H. Son, H. Lim, J.K. Ahn**, M. Kang** (2026) Sex-Specific Association of Drinking Habit and Alcohol Beverage Type with Serum Uric Acid in Health Population. *Journal of Korean Medical Science*, 41(2), e35.

Papers at Refereed Journals in 2025 (*Co-first author, **Co-corresponding author)

- Park Y. J.*, S. Lee*, S. Hong, K. Kim**, J. Kim, J. Y. Kim, K.-M. Park, Y. K. On, S.-J. Park** (2025) Temporal Trends and Prognostic Impact of Pacemaker-Associated Heart Failure: Insights from a Nationwide Cohort Study. *Journal of Clinical Medicine*, 14(21), 7744.
- Jang M.-S., W. J. Won, H. Lee, S. Hong, I. S. Lee, J. H. Lee, E. S. Ko (2025) Evaluation of Residual Disease After Neoadjuvant Chemotherapy Using Manganese-Enhanced Magnetic Resonance Imaging in an Orthotopic Mouse Model of Breast Cancer. *Investigative Magnetic Resonance Imaging*, 29(3), 184-193.
- Lim CY., Y. Seo, B. Sohn, M. Seong, S.T. Kim, S. Hong, J. Youn, E.Y. Kim (2025) The Inferior Cerebellar Peduncle Sign: A Novel Imaging Marker for Differentiating Multiple System Atrophy Cerebellar Type from Spinocerebellar Ataxia. *American Journal of Neuroradiology*, 46(6), 1223-1230.
- Lim CY., D.I. Choi, W.K. Jeong, Y.Y. Cho, S. Hong, S. Hong, K. Kim, J.-H. Kim (2025) Prediction of microsatellite-stable/epithelial-to-mesenchymal transition molecular subtype gastric cancer using CT radiomics and clinicopathologic factors. *European Journal of Radiology*, 185, 111990.

Papers at Refereed Journals in 2024 (*Co-first author, **Co-corresponding author)

- Hong S.*, S. Hong*, E. Oh, W.J. Lee, W.K. Jeong**, K. Kim** (2024) Development of A Flexible Feature Selection Framework in Radiomics-based Prediction Modeling: Assessment with Four Real-World Datasets. *Scientific Reports*, 14(1), 29297.
- Lee J. H.*, J. Shin, S*. Choi, J. Lee, S. Hong, K. Kim, J.H. Min, W.K. Jeong (2024) Preoperative Prediction of Early Recurrence in Resectable Pancreatic Cancer Integrating Clinical, Radiologic, and CT Radiomics Features. *Cancer Imaging*, 24(1), 6.

Papers at Refereed Journals in 2023 (*Co-first author, **Co-corresponding author)

- Kim M.*, M. Cho*, S. Hong, J.H. Song, E.R. Kim, S.N. Hong, D.K. Chang, Y. Kim, J.E. Kim (2023) Weight Loss from Diagnosis of Crohn's Disease to One Year Post-Diagnosis Results in Earlier Surgery. *Scientific Reports*, 13(1), 21101.

PUBLICATIONS (CONT.)**Papers at Refereed Journals in 2022** (*Co-first author, **Co-corresponding author)

- Yu J.*, S. Hong*, Y.C. Lee, K.H. Lee, I. Lee, Y. Seo, M. Kang, K. Kim, W.C. Cha, S.Y. Shin (2022) Stakeholders' Requirements of Artificial Intelligence for Healthcare in Korea. *Healthcare Informatics Research*, 28(2), 143-151.

Papers at Refereed Journals in 2021 (*Co-first author, **Co-corresponding author)

- Jang H.R.*, M. Kim*, S. Hong, K. Lee, M.Y. Park, K.E. Yang, C. Lee, J. Jeon, K.W. Lee, J.E. Lee, J.B. Park, K. Kim, G.Y. Kwon, Y.G. Kim, D.J. Kim, W. Huh (2021) Early postoperative urinary MCP-1 as a potential biomarker predicting acute rejection in living donor kidney transplantation: a prospective cohort study. *Scientific Reports*, 11(1), 1-14.
- Kwon M.J.*, J.M. Ryu*, S.Y. Cho, S.J. Nam, S.W. Kim, J. Lee, S.J. Lee, J. Park, H.Y. Park, S. Hong, K. Kim, J. Han, Y. Moon, Y.K. Shin, J.E. Lee (2021) Validation of the GenesWell BCT Score in Young Asian Women with HR+/HER2- Early Breast Cancer. *Frontiers in Oncology*, 11, 167.

Papers at Refereed Journals in 2020 (*Co-first author, **Co-corresponding author)

- Hong S., S. Lee, J. Lee, W.C. Cha**, K. Kim** (2020) Prediction of Cardiac Arrest in the Emergency Department Based on Machine Learning and Sequential Characteristics: Model Development and Retrospective Clinical Validation Study. *JMIR Medical Informatics*, 8(8), e15932.
- Lee S., S. Hong, W.C. Cha**, K. Kim** (2020) Predicting Adverse Outcomes for Febrile Patients in the Emergency Department Using Sparse Laboratory Data: Development of a Time Adaptive Model. *JMIR Medical Informatics*, 8(3), e16117.

Papers at Refereed Journals in 2019 (*Co-first author, **Co-corresponding author)

- Kim Y.*, C.H. Kim*, H.Y. Lee*, S.H. Lee*, H.S. Kim, S. Lee, H. Cha, S. Hong, K. Kim, S.W. Seo, J.M. Sun, M.J. Han, J.S. Ahn, K. Park (2019) Comprehensive Clinical and Genetic Characterization of Hyperprogression Based on Volumetry in Advanced Non-Small Cell Lung Cancer Treated With Immune Checkpoint Inhibitor. *Journal of Thoracic Oncology*, 14(9), 1608-1618.

PUBLICATION IN PROGRESS**Submitted Paper** (*Co-first author, **Co-corresponding author)

Kim J.E.*, S. Hong*, T. Shin, S. Choi, E.R. Kim, S.N. Hong, D.K. Chang, M.J. Chung, S.Y. Ha**, Y.-H. Kim**, Artificial Intelligence-Based Prediction of Nancy Histology Index using Digital Pathology in Ulcerative Colitis Patients.

Kim K.R.*, S. Hong*, S. Hong, J. Yoo, G.-J. Choi, D.R. Kim, A. Shin, N.H. Lee, H.Y. Ju, J.W. Lee, K.H. Yoo, K.W. Sung, K. Kim**, Y.-J. Kim**, Changes in Pulse Rates for Prediction of Bacteremia in Pediatric Patients with Persistent Neutropenic Fever based on Binary Prediction Models.

Working Papers

Hong S., K. Kim, Diagnostic Prediction Models in Emergency Medicine: A Review of Evaluation Metrics and Comparative Methods.

Hong S., Yuwon Kim, Ryoung-Eun Ko, S. Hong, K. Kim, Comparison of Machine Learning-Based Propensity Score Methods for Imbalanced Three Treatment Groups: Applications to Three Real-World Datasets.

POSTERS

- S. Hong, H. Choo, K.H. Lee, S. Hong, K.B. Lee, C.Y. Lee (2024) Is it beneficial to use different thresholds over time for early prediction model?, Presented at *the EFMI Special Topic Conferences 2024*, University Politehnica of Timisoara, Timisoara, Romania, November 27-29, 2024
- K.H. Lee, H. Choo, S. Hong, S. Hong, K.B. Lee, H. Cho (2024) Relationship between in-hospital sepsis prediction score and prevalence of community-onset sepsis: Triage for sepsis risk management, Presented at *the EFMI Special Topic Conferences 2024*, University Politehnica of Timisoara, Timisoara, Romania, November 27-29, 2024
- H. Choo, K.H. Lee, S. Hong, S. Hong, K.B. Lee, C.Y. Lee (2024) Beyond the ROC Curve: Activity Monitoring to Evaluate Deep Learning Models in Clinical Settings, Presented at *the EFMI Special Topic Conferences 2024*, University Politehnica of Timisoara, Timisoara, Romania, November 27-29, 2024
- J. Hwang, S. Hong, K. Kim, J. Hur, Y. Chang, J. Ahn, M. Kang (2023) Alcohol Consumption Amount, Type of Beverage, and Gender, All Matters to Serum Uric Acid Levels, Presented at *the American College of Rheumatology Convergence*, San Diego Convention Center, California, United States, November 14, 2023
- S. Hong, S. Hong, W.K. Jeong, K. Kim (2023) Development of an Efficient Feature Selection Framework in Radiomics-based Prediction Model, Presented at *the Korean Society of Health Informatics and Statistics Spring Conference*, Maria Hall 1F SONGEUI Building Catholic University, Seoul, South Korea, May 19, 2023

POSTERS (CONT.)

- S. Hong, **S. Hong**, W.K. Jeong, K. Kim (2022) Development of an Automated Framework for Optimal Feature Selection in Radiomics-based Prediction Model, Presented at *the 6th Annual SAIHST Symposium*, Hippocrates Hall 9F Irwon Station Building, Seoul, South Korea, December 21, 2022
- **S. Hong**, S. Hong, S. Lee, K. Kim, Won Chul Cha (2021) Development of Clinical Quality Monitoring System based on Common Data Model in Emergency Department, Presented at *the 5th Annual SAIHST Symposium*, Schweitzer Hall 9F Irwon Station Building, Seoul, South Korea, December 20, 2021
- S. Hong, **S. Hong**, K. Kim (2021) Impact of COVID-19 Pandemic on Occurrence Trends of Respiratory Diseases in Korea, Presented at *the Korean Society of Medical Informatics Spring Conference*, CHA Bio Complex, South Korea, July 8-9, 2021
- K.H Lee, **S. Hong**, J.Y. Yu, Y.C. Lee, I. Lee, S.Y. Shin (2020) A survey and Analysis on Healthcare AI-Project, Presented at *the Korean Society of Medical Informatics Fall Conference*, Seoul National University Siheung Campus, South Korea, November 12-13, 2020
- **S. Hong**, S. Lee, J. Lee, K. Kim, W.C. Cha (2019) Machine learning prediction of cardiac arrest in emergency department: sequential characteristics for clinical validity, Presented at *the 3th Annual SAIHST Symposium*, Hippocrates Hall 9F Irwon Station Building, Seoul, South Korea, December 23, 2019
- **S. Hong**, W.C. Cha, D. Jang, K. Kim (2018) Development of a Cardiac Arrest Prediction Model using Deep Neural Networks for the Patients in the Emergency Room, Presented at *the Korean Society of Medical Informatics Spring Conference*, Samsung Medical Center, South Korea, June 14-15, 2018

OTHER PUBLICATIONS**Book**

- Sungjun Hong, Sungsoo Hong, Don Kim, Kyu Kim, Kyungtae Kim (2022) Data Visualization and Interactive Dashboards with Python (English Language), Amazon KDP, United States.
- Hong Sungsoo, Hong Sungjun (2022) Big-Data Modeling and Visualization with QlikSense (Korean Language), BOOKK, South Korea.
- Hong Sungsoo, Hong Sungjun (2021) Big-Data Analysis and Visualization with QlikSense (Korean Language), BOOKK, South Korea.
- Hong Sungjun, Hong Sungsoo (2021) Interactive Dashboards with Plotly & Dash (Korean Language), BOOKK, South Korea.

Report

- Sungjun Hong, Kyunga Kim, Won Chul Cha (2021) KHIDI Digital Healthcare Report: Management System of Big Data-based Emergency and Intensive Care Unit (Korean Language), Korea Health Industry Development Institute.

Patent

“Radiomics Features Selection Method for Artificial Intelligence Model and Analysis Apparatus”, Woo Kyoung Jeong, Kyunga Kim, Sungsoo Hong, Sungjun Hong, Korean Patent No. 10-2023-0133813 (2023)

Full List of Research Projects

RESEARCH PROJECTS

- Development of Multimodality AI Model for Predicting Patterns of Visual Field Defect in Glaucoma Patients (Jan. 2024 – Feb. 2025)
 - Samsung Medical Center (PI: Eun Jung Lee; Ophthalmology @ SMC)
 - Applied clustering methods to identify patterns of visual field defects in glaucoma patients (Python)
 - Developed a prediction model for visual field defects in glaucoma patients
- Development of Prediction Model for Post-gastrectomy Complication (Jan. 2024 – Feb. 2025)
 - Samsung Medical Center (PI: Sung Eun Oh; Gastric Cancer Center @ SMC)
 - Conducted risk factor analysis for complications in post-gastrectomy patients (R)
 - Developed a prediction model for complication in post-gastrectomy patients
- Investigating the Usefulness of Histological Activity Evaluation using Deep Learning in Patients with Ulcerative Colitis (Sep. 2023 – Dec. 2024, Jun. 2025 – Mar. 2026)
 - Samsung Medical Center (PI: Young-Ho Kim; Gastroenterology @ SMC)
 - Developed a segmentation model for multiple cell types and a prediction model for Nancy-grade activity in patients with UC (Python)
 - Conducted external validation using a multi-center dataset
- Development of Bacteremia Prediction Model in Pediatric Cancer Patients with Persistent Neutropenic Fever (Jan. 2022 – Aug. 2023, Nov. 2025 – May 2026)
 - Samsung Medical Center (PI: Yae Jean Kim; Pediatric Adolescent Center @ SMC)
 - Conducted risk factor analysis to identify predictors of bacteremia
 - Developed a prediction model for bacteremia in pediatric cancer patients (R)
 - Conducted temporal validation for the prediction model
- Converged Image-based Precision Oncology AI Solution for Gastrointestinal Malignancies (May 2021 – Aug. 2023)
 - Samsung Medical Center, Future Medicine 20*30 Project (PI: Woo Kyoung Jeong; Radiology (Gastric Cancer Center) @ SMC)
 - Designed and validated a clinical prediction model for radio-genomics
 - Developed a radiomic-based feature selection framework (R)
- A Comparative Study of the Correlation between Insulin Resistance or Diabetes and Dietary Patterns applying Machine Learning and Statistical Techniques (Sep. 2022 – Feb. 2023)
 - PI: Yoong Jung Yang; Department of Food and Nutrition @ DWU, Kyunga Kim; Biomedical Statistics Center @ SMC
 - The Korea national health and nutrition examination survey
 - Applied clustering methods to identify dietary intake patterns (R, Python)
- A Study on Non-parametric Propensity Score Method in Unbalance Multiple Treatment Groups (Jun. 2021 – Feb. 2023)
 - National Research Foundation of Korea (No. 2021R1F1A1054806, PI: Kyunga Kim; Biomedical Statistics Center @ SMC)
 - Developed statistical methodologies for PS matching in multiple-treatment settings, and evaluated performance using both simulation studies and real-world datasets (R)

RESEARCH PROJECTS (CONT.)

- Development and Validation of Serum Uric Acid Levels in Prediction Model using Health Promotion Center Data (Mar. 2021 – Jun. 2022)
 - Samsung Medical Center (PI: Mira Kang; Health Promotion Center @ SMC)
 - Developed a prediction model of serum uric acid levels using health check-up data
 - Conducted risk factor analysis to evaluate the association between alcohol intake and serum uric acid levels (R)
- Development and Implementation of CDM, FHIR and AI-based Precision Medicine Platform to improve National Medical Quality and Safety for Emergency and Critical Patients (Apr. 2019 – Dec. 2021)
 - Ministry of Health and Welfare (No. HI19C0275, PI: Won Chul Cha; Emergency Medicine @ SMC)
 - Part II. Conducted qualitative research for hospital quality indicators based on CDM and developed a management dashboard program (PI: Kyunga Kim; Biomedical Statistics Center @ SMC)
- The Effect of CRT Upgrade to Overall Survival in Pacing-Associated Heart Failure Patients (Aug. 2020 – Dec. 2023)
 - Samsung Medical Center (PI: Seung Jung Park; Heart Vascular Stroke Institute @ SMC)
 - Designed and analyzed Korea National Health Insurance claims data
 - Conducted survival analysis to evaluate the incidence of heart failure following pacemaker implantation (R, SAS)
 - Performed subgroup analysis using propensity score matching and time-dependent modeling (time-varying covariates and coefficients)
- Development and validation of a score model for predicting time-dependent mortality in very low birth weight Infant (VLBWI) (Jun. 2020 – Mar. 2021)
 - National Research Foundation of Korea (No. 2020R1F1A107581711, PI: Myung Hee Lee; Biomedical Statistics Center @ SMC)
 - Developed and validated a regression-based clinical risk scoring system (R)
- Implementation of Prediction Model for Cardiac Arrest in Samsung Medical Center Emergency Department (Nov. 2020 – Feb. 2021)
 - Samsung Medical Center (PI: Won Chul Cha; Emergency Medicine @ SMC)
 - Applied a prediction model for cardiac arrest in ER (Python)
- Strategy Planning of Artificial Intelligence for Public Health and Medicine (Jul. 2020 – Nov. 2020)
 - Korea Health Information Service (PI: SooYong Shin @ SKKU)
 - Conducted expert interviews and analyzed survey data to inform strategic planning
- Development and Validation of Prediction Model for Cardiac Arrest in Samsung Medical Center Emergency Department (Jan. 2018 – Jul. 2019)
 - Samsung Medical Center (PI: Won Chul Cha; Emergency Medicine, Kyunga Kim; Biomedical Statistics Center @ SMC)
 - Developed and validated a prediction model for cardiac arrest risk (R, Python)

RESEARCH PROJECTS (CONT.)

- Clinical Performance Study of Breast Cancer Prognostic Diagnostics GenesWell™ BCT in Korean Patients with Hormone Receptor-positive, Human Epidermal Growth Factor Receptor 2-negative Early Breast Cancer (Nov. 2017 – Feb. 2020)
 - Ministry of Health and Welfare (PI: JeongEon Lee; Breast Cancer Center @ SMC)
 - Conducted survival analysis to compare outcomes between high- and low-risk groups and evaluate clinicopathologic characteristics based on GenesWell™ BCT values (R)

- Estimation of Usual Intake with Consideration of Within-Person Variance (Mar. 2018 – Nov. 2018)
 - Ministry of Food and Drug Safety (PI: JungEun Lee; Department of Food and Nutrition @ SNU)
 - Conducted simulation studies to evaluate statistical methods for estimating usual dietary intake (R)

- A Study on the Job Administration Statistics by Small Area (Aug. 2017 – Nov. 2017)
 - Statistics Korea (PI: Yongil Jeon @ SKKU)
 - Developed algorithms to estimate labor and employment statistics (SAS)

REFERENCES

Seohyeon Park, Ph.D.

Postdoctoral Fellow, Department of Population Health

NYU Langone Health

Email: Seohyeon.park@nyulangone.org

Kyunga Kim, Ph.D.

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